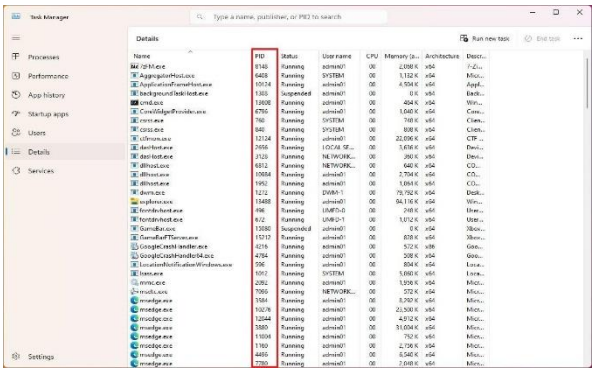
Windows basics: processes, services, registry, users and groups, UAC, and the three main Event Viewer logs (Application, Security, System)

1. Windows Processes

A process is an instance of an executable program running in memory.



A process may contain:

- Process ID (PID).
- Parent Process ID (PPID).
- Executable path.
- User account context.
- Security token.
- Threads.
- Memory and system resources.

Important Windows processes:

Process	Function
explorer.exe	Windows graphical shell and File Explorer
services.exe	Service Control Manager
lsass.exe	Local Security Authority Subsystem Service
svchost.exe	Hosts Windows services
winlogon.exe	Handles interactive logon-related operations
csrss.exe	Client/Server Runtime Subsystem
smss.exe	Session Manager Subsystem

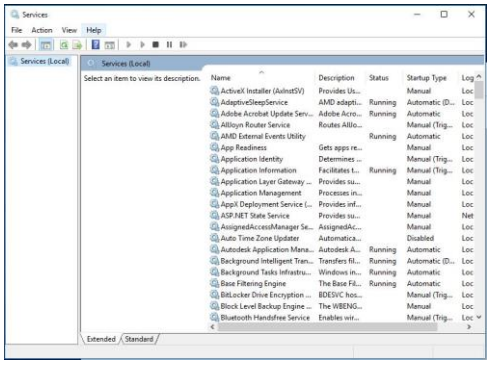
Security analysis focuses on suspicious executable paths, unusual parent-child process relationships, unexpected command lines, and unauthorized execution.

Useful command:

Get-Process

2. Windows Services

A Windows service is a background program managed by the Service Control Manager.



Services can start automatically, manually, or in response to system events.

They may run under accounts such as LocalSystem, LocalService, NetworkService, or configured user accounts.

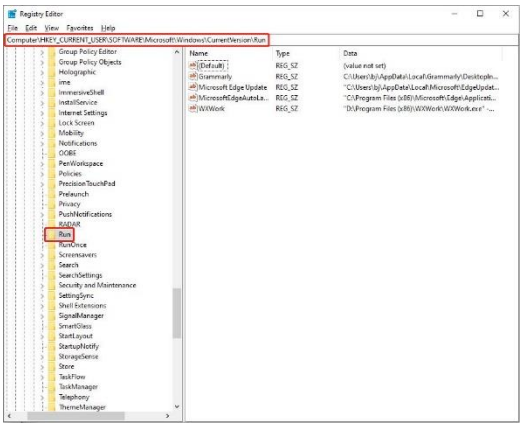
Useful command:

Get-Service

Security relevance: Malicious services can be used for persistence, privilege abuse, and execution of unauthorized programs.

3. Windows Registry

The Windows Registry is a hierarchical database that stores operating system and application configuration.



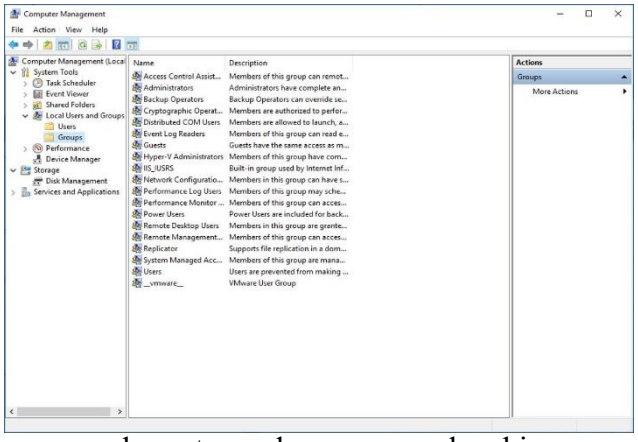
Important registry hives:

Registry Hive	Purpose
HKLM	Machine-wide settings
HKCU	Current user's settings
HKU	Loaded user profile settings
HKCR	File associations and COM configuration
HKCC	Current hardware profile settings

A common user-level startup location is:
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run

Registry analysis helps identify persistence mechanisms, suspicious configuration changes, and unauthorized startup programs.

4. Windows Users and Groups



Windows uses user accounts and security groups to manage authentication and authorization.
Important groups include:

- Administrators.
- Users.
- Guests.
- Remote Desktop Users.
- Backup Operators.

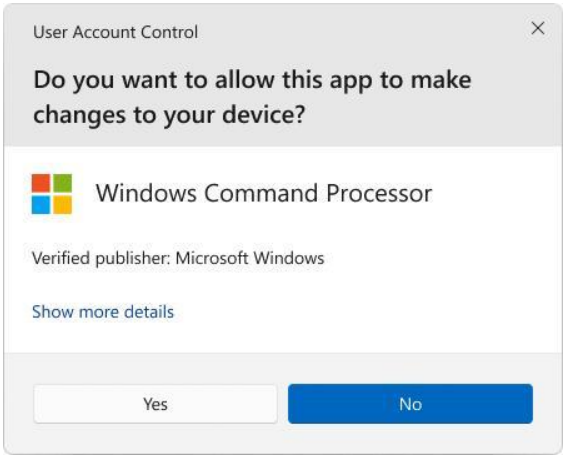
Security monitoring includes detecting unauthorized account creation, privilege changes, password resets, and group membership modifications.

Useful commands:

whoami
whoami /all
Get-LocalUser

Get-LocalGroup

5. User Account Control (UAC)



User Account Control (UAC) is a Windows security feature designed to control administrative elevation.

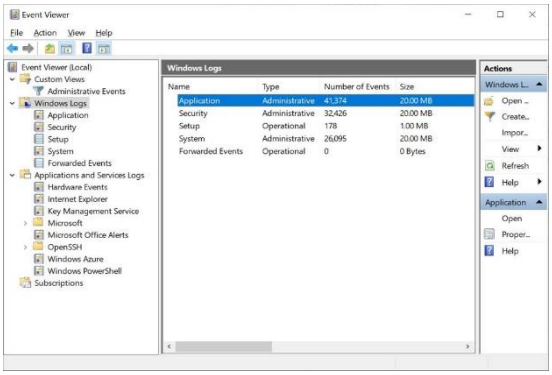
It helps prevent applications from automatically performing privileged operations without appropriate authorization.

UAC can require administrator consent or credentials before an operation receives elevated privileges.

Its main objectives include reducing unintended administrative changes and encouraging applications to operate with standard user privileges.

UAC is not a complete security boundary and does not prevent every privilege escalation attack.

6. Windows Event Viewer



Event Viewer is a Windows management tool used to examine operating system, application, and security event logs.

It can be opened using:

eventvwr.msc

Three Main Event Viewer Logs

Log	Description	Security Importance
Application	Records application-generated events, including errors and warnings	Identifies abnormal application behavior
Security	Records audited authentication, authorization, and account-management events	Supports intrusion detection and forensic investigation
System	Records events from Windows components, drivers, and services	Helps identify service changes, system failures, and restarts